IT4010 Research Project – 2025 July Topic Assessment Form

Project ID : 25-26J-292

1. Topic (12 words max)

AI-Powered Tea Quality and Moisture Prediction with Intelligent Auction Price Forecasting

2. Research group the project belongs to

CoEAI - Centre of Excellence for AI

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

Sri Lanka's tea industry is internationally acclaimed for its high-quality produce, contributing significantly to the nation's economy. However, despite its global reputation, many core operational areas—particularly tea grading and evaluation—still rely heavily on manual, experience-driven, and often subjective methods. At present, expert tea tasters visually examine dried tea samples and provide qualitative comments that influence grading and, ultimately, auction outcomes [3]. While their evaluations are based on years of experience, this process is time-consuming, varies between individuals, and can introduce inconsistencies across batches. Such subjectivity can lead to mis-grading, reduced buyer confidence, and inefficiencies in maintaining product quality across large-scale operations [1].

Furthermore, in the context of modern supply chain demands and global competition, scalability and standardization are critical. However, the current human-centric tea grading method makes it difficult to scale evaluations without compromising on accuracy. This becomes especially problematic when managing large volumes of different tea grades such as Orange Pekoe (OP), Broken Orange Pekoe One (BOP1), and Orange Pekoe (OPA), where visual characteristics like leaf size, twist, and uniformity are key differentiators. The absence of a consistent, data-driven evaluation method restricts the industry's ability to ensure quality assurance and transparency at scale [1], [4].

In parallel, the quality of tea is also affected at the very beginning of the production cycle—during leaf plucking. The standard practice is to harvest two leaves and a bud, but in real-world conditions, ensuring every tea Plucker adheres to this rule is challenging.

Over-plucking or under-plucking directly affects the chemical composition and physical structure of the leaves, degrading final product quality. With large labor forces in tea estates, real-time monitoring and feedback on Plucker performance is not feasible using traditional methods, resulting in quality variability that is hard to trace and control [2].

To address these limitations, this research introduces a two-pronged AI-based solution. The first component is an automated tea grading and comment generation system that uses machine learning techniques—including classical ML, CNNs, and hybrid models—to classify specific tea types (OP, BOP1, FBOP) based on their visual characteristics. The system also generates human-like descriptive comments to replicate expert tasters' insights, making the results interpretable and suitable for both evaluation and training purposes [1], [4]. The second component is a vision-based plucking stage detection module, which uses real-time image analysis of fresh tea leaves (e.g., from a conveyor belt system) to categorize them as under-plucked, correctly plucked, or over-plucked. These classifications are logged against worker identifiers to assess individual performance and enable targeted feedback [2].

Existing research has yet to integrate these two domains—grading and plucking—into a cohesive system that not only automates key processes but also augments human expertise through explainable AI. The goal of this study is to improve efficiency, reduce subjectivity, and enhance quality assurance throughout the tea production pipeline. By aligning with global expectations for traceability and standardization, this project aims to contribute to a more competitive and technologically empowered Sri Lankan tea industry [3].

References

[1] T. Ghosh, A. Das, and A. Roy, "Tea leaf grading using image processing and machine learning," International Journal of Computer Applications, vol. 179, no. 41, pp. 1–5, 2018.

[2] K. Subasinghe and R. Madurapperuma, "A computer vision-based plucking point identification system for tea," Journal of the National Science Foundation of Sri Lanka, vol. 48, no. 3, pp. 317–327, 2020.

[3] Ceylon Chamber of Commerce, "The Colombo Tea Auction." [Online]. Available: https://www.chamber.lk

[4] S. Gunaratne, A. Bandara, R. Liyanage, and I. Perera, "Quality assessment of black tea using computer vision and deep learning," IEEE Access, vol. 9, pp. 76532–76541, 2021.

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

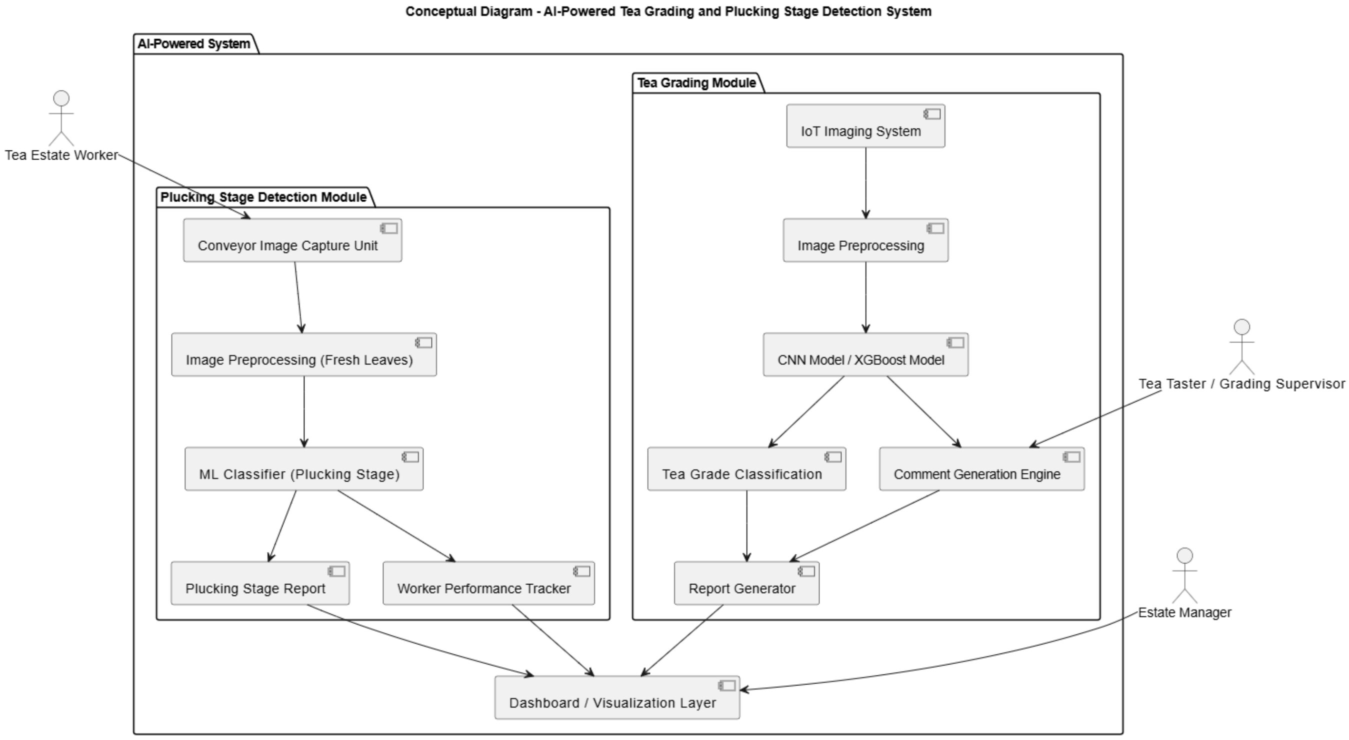
This research introduces an integrated AI-based system designed to enhance the efficiency, accuracy, and standardization of tea quality evaluation processes in Sri Lanka. The proposed solution consists of two primary modules: a Tea Grading Module and a Plucking Stage Detection Module, both supported by IoT infrastructure and machine learning techniques.

The Tea Grading Module captures high-resolution images of dried tea leaves (OP, BOP1, OPA, etc.) using a custom-designed IoT conveyor belt system under controlled lighting. These images are processed and classified using deep learning models (CNN, XGBoost), which extract visual features like leaf shape, twist, evenness, and color. A comment generation engine provides human-like descriptions to replicate expert tasters' feedback, aiding in both automated evaluation and training.

The Plucking Stage Detection Module uses a separate conveyor-based imaging setup to analyze fresh tea leaves. Images are processed in real-time to classify leaves as under-plucked, correctly plucked, or over-plucked using ML models. Results are linked to individual worker IDs to enable performance monitoring and targeted feedback.

Both modules are integrated into a unified dashboard that visualizes grading results, plucking quality insights, and performance analytics for estate managers. This system reduces subjectivity, improves traceability, and offers a scalable, interpretable solution aligned with modern agricultural digitization goals.

A conceptual diagram is provided to illustrate the system's architecture, highlighting hardware components, AI models, and stakeholder interactions.



7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

This research requires expertise from both the tea industry and machine learning domains to ensure practical accuracy and relevance. For the tea grading component, the involvement of professional tea tasters or grading supervisors is essential. Their insights will help define key attributes such as leaf size, shape, twist, texture, color uniformity, and stalk content, which influence grading decisions in tea auctions. These experts will also support the evaluation of AI-generated taster-style comments, ensuring that the outputs align with real-world tasting language commonly used by industry professionals.

For the plucking stage detection component, input from tea estate supervisors and field-level managers is vital. Their domain knowledge will support the proper labeling of leaf images as under-plucked, correctly plucked, or over-plucked based on accepted agricultural standards. These experts can also provide guidance on how plucked leaves should be captured—ideally on a conveyor belt or collection tray under consistent lighting—to simulate real estate or factory conditions. This will enhance the reliability of the plucking stage classification system and help connect results to worker performance evaluations.

In terms of data requirements, the project demands high-quality, expert-labeled image datasets. For grading, images of dried tea leaves across different grades must be collected under varied lighting and batch conditions. For plucking stage analysis, fresh leaf images from estate environments or controlled simulations must be categorized accurately. Additionally, a repository of professional tea taster comments is required to support the development of rule-based or template-driven comment generation models. Domain expertise will be essential in annotating these datasets and validating system outputs, ensuring the research results are both academically valid and practically applicable within the tea industry.

8. Objectives and Novelty

Main Objective			
To develop an AI-based system that automates the tea grading process and plucking stage detection using image processing and machine learning techniques, with the aim of improving evaluation accuracy, standardizing quality assessment, and supporting tea taster training and Plucker performance analysis in the Sri Lankan tea industry.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
IT22223326 - De Silva P. S	To develop a computer vision-based model that emulates human tea taster evaluation by analyzing visual properties of tea leaves.	• Design and construct a light-controlled imaging setup with a 45-degree camera angle to capture dry and wet tea leaf samples in consistent conditions. • Collect image data across diverse tea samples with known taster scores, capturing both dry leaf and infused (wet) leaf views. • Extract features related to visual quality indicators like color depth, twist shape, bloom, and	• Mimics expert tea tasters' evaluation by capturing subtle visual cues, enabling digitization of a highly subjective process. • Incorporates both dry and infused leaf assessments—a rarely integrated dual-view method in automated tea grading. • Uses controlled lighting and angled macro photography to simulate the human sensory evaluation environment.

		uniformity using image processing techniques . • Train a deep learning model (CNN) to classify tea leaf quality based on visual features and generate corresponding taster-style scores (e.g., 1–10 scale). • Integrate a scoring system to reflect the standard sensory evaluation process followed by professional tasters. • Validate model accuracy using expert tea taster feedback and measure consistency against human evaluations. • Build a simple GUI or scoring tool to visualize sample evaluation results in real-time.	• Translates subjective taster feedback into quantifiable ML-based scoring, supporting standardization in quality control. • Contributes to digitized quality grading in the tea industry, promoting consistency, speed, and scalability across estates.
IT22224248 - Shavinda G.M. V	To detect and classify the tea plucking stage (under-plucked,	• Design a conveyor-based image capture system. • Collect and label images of fresh tea leaves.	• Introduces a real-time, image-based ML system for classifying tea plucking stages (under-, correct-, and over-plucked),

	correctly plucked, over-plucked) using image processing and machine learning.	• Develop an ML model to classify plucking stages. • Link results to worker IDs for performance tracking. • Store and analyze Plucker efficiency data.	replacing subjective manual inspection. • Uses a conveyor-mounted camera and visual cues (bud-to-leaf ratio, size, color, edge sharpness) to ensure consistent plucking assessment. • Links plucking quality to worker IDs for automated performance tracking and feedback. • Enables data-driven workforce optimization and lays the foundation for predictive analytics in estate management.
IT22254702 - Sooriyaarachchi N. D	To develop a hybrid moisture prediction system that combines real-time NIR sensor data with local weather forecasts to estimate the pre-fermentation moisture content	• Integrate portable NIR sensor for capturing absorption spectra of fresh tea leaves. • Use a weather API to gather real-time and short-range forecast data (humidity, temperature, wind, rainfall).	• Combines multimodal data: fusing spectral leaf properties with dynamic environmental factors for higher prediction accuracy. • Enables pre-fermentation decision-making, reducing the risk of poor-quality fermentation outcomes due to inaccurate moisture estimation.

	of tea leaves and support optimal fermentation timing decisions.	• Collect moisture-labeled training samples by correlating NIR + weather inputs with actual lab-tested moisture percentages. • Engineer features from both data sources (e.g., average NIR absorbance bands, humidity index, recent rainfall). • Train regression models (LightGBM, Random Forest, or Neural Networks) to predict leaf moisture levels. • Validate the model using cross-validation and unseen test samples. • Develop a front-end interface (mobile or desktop) that displays predicted moisture and gives recommendations (e.g., "Ready for fermentation", "Too moist – delay").	• Operates in real-time with field-level deploy ability through mobile or IoT devices. • Avoid destructive sampling methods or expensive drying apparatus, using non-invasive NIR and accessible weather APIs. • Supports climate-adaptive processing by accounting for rapid environmental shifts that affect leaf moisture retention.

IT22211996 - Liyanage S. N	To develop a machine learning-based model for predicting the auction price of Tea	• Design and build a data-driven tea auction prediction system integrating structured and environmental data from tea factories and auction records. • Collect and preprocess structured datasets containing tea grade, region, auction date, season, leaf appearance indicators, weather conditions (rainfall and temperature), volume sold, and historical auction prices. • Engineer features from temporal, geographical, and environmental variables, and encode categorical data to ensure model readiness. • Use a hybrid machine learning approach combining time-aware features and XGBoost regression to predict auction prices based on multivariate inputs. • Apply feature selection and regularization techniques to	• Introduces a structured, multi-domain ML pipeline combining auction, weather, and grading data for price prediction. • Leverages XGBoost for interpretable forecasting, revealing key economic and environmental drivers of auction prices. • Applies SHAP-based explainability to offer transparent insights supporting strategic decisions in tea marketing. • Bridges data science and agriculture by forecasting tea prices using real-world, region-specific variables.

		reduce noise and enhance model interpretability. • Evaluate model performance using cross-validation metrics (RMSE, MAE) and interpret the impact of each feature using SHAP values and partial dependence plots. • Visualize predicted price trends and build a user-friendly decision-support dashboard to assist tea stakeholders in pricing strategy and market forecasting.	

9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
IT22223326 - De Silva P. S	This component aims to replicate the visual evaluation process of an expert tea taster using image processing and machine learning. It focuses on key sensory indicators such as leaf color, tip presence, twist, bloom, brightness, and infused leaf appearance to assess tea quality. By capturing both dry and wet leaf samples in a controlled imaging setup, the system learns to quantify visual tea attributes and translate them into quality scores. The goal is to build a scalable, objective tool that supports consistent tea grading, assists in quality assurance, and reduces reliance on subjective human assessments in Sri Lanka's tea industry.
IT22224248 - Shavinda G.M. V	The task involves designing and implementing an automated system for detecting and classifying tea plucking stages using machine learning and image processing techniques. This includes developing a conveyor-based image capture setup to consistently acquire high-quality images of fresh tea leaves under controlled conditions. The collected images are labeled to create a robust dataset for supervised learning. A machine learning model is developed to accurately classify plucking stages such as under-plucked, correctly plucked, and over-plucked. The system integrates worker identification to link classification results with individual pluckers, enabling real-time performance tracking. Additionally, the solution stores and analyzes plucker efficiency data, supporting workforce management and continuous quality improvement. The entire development follows a structured software lifecycle, from system design to testing and validation, ensuring practical deployment and adherence to best practices.
IT22254702 - Sooriyaarachchi N. D	This component focuses on predicting the moisture percentage of tea leaves before fermentation by fusing spectral data from NIR scans with local weather data. Unlike traditional moisture measurement methods that require drying or lab testing, this system enables on-the-field, real-time predictions. It considers recent and forecasted environmental conditions (humidity, temperature, rainfall) along with internal leaf moisture indicators from NIR absorption spectra to model the moisture dynamics of plucked leaves. The goal is to prevent premature or delayed fermentation based on inaccurate moisture levels, improving consistency in tea quality. The prediction model can be deployed on mobile or embedded platforms for factory or estate-level decision-making.

IT22211996 - Liyanage S. N	This component focuses on building a system to predict tea auction prices based on historical data, tea grades (e.g. BOP1), regions, and weather conditions. The system uses supervised learning techniques to analyze patterns and forecast future prices. It supports tea producers and exporters in making informed decisions by providing accurate and timely price predictions. A visual dashboard displays trends and insights, making the results easy to interpret. The component follows a structured development process, ensuring accuracy, transparency, and real-world applicability within the Sri Lankan tea industry's economic forecasting and planning needs.

10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor				
Co-Supervisor				
External Supervisor				
Summary of external supervisor’s (if any) experience and expertise				

<u>This part is to be filled by the Topic Screening Staff members.</u>

a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes		No	

b) Does the proposed topic exhibit novelty?

Yes		No	

c) Do you believe they have the capability to successfully execute the proposed project?

Yes		No	

d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes		No	

e) Supervisor's Evaluation and Recommendation for the Research topic:

IT4010 Research Project – 2025 July Topic Assessment Form

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

*__Important__:

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.